Code: 20A03703c **R20**

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

GEOMETRIC DIMENSIONING AND TOLERANCES

(Mechanical Engineering)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	Define datum axis, datum target lines, datum target areas, datum target point representation.	2M
	(b)	Define, the types of dimensioning.	2M
	(c)	Define unit flatnes, perfect flatness.	2M
	(d)	What are the principles of dimensioning?	2M
	(e)	Define runout and types of runouts in geometrical tolerance.	2M
	(f)	Define any 4 categories, characteristics and symbols of geometric Characteristic Symbols.	2M
	(g)	What is meant by filtering in surface roughness measurement?	2M
	(h)	Define Ra, Rz, Rp, Rv, Rsk.	2M
	(i)	Mention various types of profilometers.	2M
	(j)	What is meant by measurement uncertainty?	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2	(a)	Define primary, Secondary and tertiary datum planes along with datum target areas with diagram.	5M
	(b)	Explain 3 – 2 -1 principle with diagram.	5M
		OR	
3	(a)	Elaborate degree of freedom with a example.	5M
	(b)	Explain the basic principles of rule1 and rule 2 with an example.	5M
4		Define radial element and describe how a radial element specification is applied to a drawing. Explain with a minimum of two examples.	10M
		OR	
5		Define tangent plane and describe the relationship between the actual surface, the tangent plane, and the geometric tolerance.	10M
6		Define positional Tolerancing of Nonparallel Holes with two examples.	10M
		OR	
7		Define fundamental dimensioning rules applying to any of the two industrial applications parts.	10M
8		Explain 2D-measurement method of surface roughness (i) on cylindrical surface, (ii) on linear surface.	10M
		OR	
9		In the measurement of surface roughness, heights of 12 successive peaks and valleys were measured from a datum as Peaks- 41, 43, 40, 33, 35, 32 microns Valleys 32, 25, 26, 24, 19, 42 microns Determine the Ra Value.	10M
10		Differentiate conventional and CMM method of inspection with an example.	10M
		OR	
11		With an example, Explain in detail about how contact type profilometer works.	10M

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PART – A

(Compulsory Question)

*****

1 Answer the following: (10 X 02 = 20 Marks)

(a)	Define limits.	2M
(b)	What are the applications of simulators?	2M
(c)	Give a brief note on principles of dimensioning.	2M
(d)	List the steps involved in measuring the roundness.	2M
(e)	Define runout.	2M
(f)	What is location measurement?	2M
(g)	Differentiate between primary and secondary texture?	2M
(h)	What are the methods to measure Surface finish?	2M
(i)	What is statistical tolerancing?	2M
(j)	List the applications of CMM.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2 What is datum? Explain the role of datum systems. 10M

OR

3 Explain in detail the procedure involved in the inspection of dimensional and geometrical deviations. 10M

4 What is form tolerance? Explain with suitable examples. 10M

OR

5 Explain various types of orientation tolerances. Also explain their specifications and interpretations. 10M

6 Explain various types of profile tolerances. Also explain their specifications and interpretations. 10M

OR

7 Discuss in detail the verification techniques. Give suitable examples. 10M

8 How is surface roughness calculated by CLA and R.M.S methods? 10M

OR

9 The heights of peaks and valleys of 20 successive points on a surface are 35, 25, 40, 22, 37, 19, 41, 21, 42, 18, 42, 24, 44, 25, 40, 18, 40, 18, 39, and 21 microns respectively, measured over a length 20 mm. Determine CLA and RMS values of roughness surface. 10M

10 Write a detailed note on computer aided tolerancing and verification. 10M

OR

11 Explain the following: 10M

(i) Working principle of CMM.

(ii) Dimensional Chains.

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